Thomas Holterbach

Co-founder of bgproutes.io
PhD in Computer Networks, ETH Zurich

✉ *thomasholterbach@gmail.com*
French, born on 31 July 1991



Education

Doctoral degree in Computer Science, ETH Zürich, Switzerland — Aug. 2021
Advised by Prof. Laurent Vanbever in the Networked Systems Group
- Passed the Machine Learning (8 credits) & Data Mining (4 credits) courses

Master degree in Computer Science, University of Strasbourg, France — June 2014
Specialized in Networks and Embedded systems
With honors. Ranked 2'/15. Grade: 15.2/20

Bachelor degree in Computer Science, University of Strasbourg, France — June 2012
Ranked 5'/50. Grade: 13.87/20

Professional Experience

Co-founder of bgproutes.io, France (with Dr. Thomas Alfroy) — May 2026 - Present
- bgproutes.io runs tools for deeper visibility into BGP routing.
- bgproutes.io also provides a public platform that collects BGP data from hundreds of real networks and serves hundreds of network operators each week.
- I created a YouTube series on BGP monitoring: *youtube.com/@bgproutes_io*.

Postdoctoral researcher, University of Strasbourg, France — Sept. 2021 - May 2026
Advised by Prof. Cristel Pelsser in the Computer Networks Group
- I designed new algorithms to compress BGP data.
- I developed a system to quickly detect forged-origin hijacks.

The system is open source and runs at *dfoh.uclouvain.be/*.

Postdoctoral researcher and Instructor, Georgia Tech, GA, USA — Sept. 2022 - Dec. 2022
Advised by Prof. Alberto Dainotti in the Internet Intelligence Lab
- I researched how to build a data-driven network Telescope.
- I was an instructor in a Master-level class about Internet routing and P4.

Teaching Assistant, ETH Zürich, Switzerland — Feb. 2022 - Apr. 2022
Advised by Prof. Laurent Vanbever in the Networked Systems Group
- I operated the "mini-Internet" platform in a class with >150 students.

The platform is open source (*mini-inter.net*) and used in many universities world-wide.

Research and Teaching Assistant, ETH Zürich, Switzerland — Feb. 2015 - Aug. 2021
Supervised by Prof. Laurent Vanbever in the Networked Systems Group

- I developed a framework to improve the Internet convergence upon failures. Published in [2,3,6], see *blink.ethz.ch* and *swift.ethz.ch* for further information.
- I developed a "mini-Internet" platform to give students hands on experience routers configuration. We published the platform in [5].

Visiting Student, University College London, UK — May 2018 - July 2018
Supervised by Dr. Stefano Vissicchio in the Department of Computer Science

- I developed a P4 program to detect Internet failures in the data plane. The program is used in [3], and publicly available at *github.com/nsg-ethz/Blink*.

Junior Researcher, University of California San Diego, CA, USA — June 2016 - Dec. 2016
Supervised by Dr. Alberto Dainotti in the Center for Applied Internet Data Analysis

- I measured the Internet convergence time upon failures using real BGP data as well as a simulator. Results are used in [2].

Research Intern, Internet Initiative Japan, Tokyo, Japan — June 2014 - Dec. 2014
Supervised by Prof. Cristel Pelsser and Randy Bush

- I studied how accurate are Internet measurement platforms such as RIPE Atlas, and we published the work in [1].

Research Intern, University of Strasbourg, France — June 2013 - Aug. 2013 & Jan. 2014 - July 2014
Supervised by Dr. Pascal Merindol in the ICube laboratory

- I optimized algorithms designed to prevent forwarding loops. Results are presented in my Master thesis: *dropbox.com/s/fuvamzi5sv1hawq/Mthesis.pdf*.

Research Publications

Papers in Top International Conferences

[12] A System for Fast, Accurate, and Intuitive Querying of Large-Scale BGP Datasets in IMC 2026
T. Holterbach, T. Alfroy, A. Mohsenpour, T. Krenc, K. C. Claffy, and C. Pelsser

[11] MORP4: A Dynamic Network Telescope in NSDI 2026
I. Xygkou, J. K. Sojan, D. Rauthan, F. Zhu, T. Holterbach, S. Alcock, B. Flanagan, A. Saeed, and A. Dainotti

[10] The Next Generation of BGP Data Collection Platforms in SIGCOMM 2024
🏆 Best Paper Award
T. Alfroy, T. Holterbach, T. Krenc, KC Claffy, and C. Pelsser

[8] A System to Detect Forge-Origin BGP Hijacks in NSDI 2024
T. Holterbach, T. Alfroy, A. Phokeer, A. Dainotti and C. Pelsser

[7] Internet Science Moonshot: Expanding BGP Data Horizons, in HotNets 2023
T. Alfroy, T. Holterbach, T. Krenc, KC Claffy, C. Pelsser

[5] An Open Platform to Teach How the Internet Practically Works in CCR 2020
🏆 One of the three "Best of CCR" papers presented at SIGCOMM 2020
T. Holterbach, T. Bühler, T. Rellstab and L. Vanbever

[4] (Self) Driving Under the Influence: Intoxicating Adversarial Network Inputs, in HotNets 2019
R. Meier, T. Holterbach, S. Keck, M. Stähli, V. Lenders, A. Singla and L. Vanbever

[3] Blink: Fast Connectivity Recovery Entirely in the Data Plane in NSDI 2019
T. Holterbach, E. Costa Molero, M. Apostolaki, S. Vissicchio, A. Dainotti and L. Vanbever

[2] SWIFT: Predictive Fast Reroute in SIGCOMM 2017
T. Holterbach, S. Vissicchio, A. Dainotti and L. Vanbever

[1] Quantifying Interferences Between Measurements on the RIPE Atlas Platform in IMC 2015
T. Holterbach, C. Pelsser, R. Bush and L. Vanbever

Posters, Demos, and Technical Reports

[9] Measuring Internet Routing from the Most Valuable Points in arXiv 2024
T. Alfroy, T. Holterbach, T. Krenc, KC Claffy, and C. Pelsser

[6] A Framework To Fast Reroute Traffic Upon Remote Outages, Doctoral Thesis, 2021
Available at *research-collection.ethz.ch/handle/20.500.11850/521609*
T. Holterbach

MVP: Measuring Internet Routing from the Most Valuable Points in IMC 2022
T. Alfroy, T. Holterbach and C. Pelsser

Boosting the BGP convergence in SDXes with SWIFT, in SIGCOMM 2017
P. Mao, R. Birkner, T. Holterbach and L. Vanbever

Measurement Vantage Point Selection Using A Similarity Metric in IRTF & ISOC ANRW
T. Holterbach, E. Aben, C. Pelsser, R. Bush, L. Vanbever

Supercharge me: Boost Router Convergence with SDN, in SIGCOMM 2015
M. Alan Chang, T. Holterbach, M. Happe, L. Vanbever

Presentations at Operational Conferences

Expanding Routing Visibility at IXPs with BMP — Oct. 2026
RIPE 93, Sofia, Bulgaria

Expanding Routing Visibility at IXPs with BMP — Sep. 2026
BalticNOG, Riga, Latvia

Expanding Routing Visibility at IXPs with BMP — Sep. 2026
European Peering Forum, Paris, France

How bgproutes.io supports latest technologies such as RPKI ROV, ASPA, and BMP — Apr. 2026
RIPE SEE, Belgrade, Serbia

bgproutes.io: A next-gen BGP data collection platform — Apr. 2026
SwiNOG, Bern, Switzerland

The Next Generation of BGP Data Collection Platforms — Feb. 2026
APRICOT, Jakarta, Indonesia

The Next Generation of BGP Data Collection Platforms — Jan. 2026
CSNOG, Zlín, Czech Republic

Expanding Internet Routing Visibility with BMP — Nov. 2025
MENOG, Dubai, United Arab Emirates

Étendre la visibilité du routage BGP grâce à BMP Oct. 2025
FRNOG, Paris, France

Expanding Internet Routing Visibility with BMP Sep. 2025
NLNOG, Amsterdam, Netherlands

The Next Generation of BGP Data Collection Platforms Jun. 2025
NANOG 94, Denver, USA

The Next Generation of BGP Data Collection Platforms May 2025
RIPE 90, Lisbon, Portugal

La prochaine génération de plateformes de collecte de données BGP Sep. 2024
FRNOG, Paris, France

A System to Detect Forge-Origin Hijacks Feb. 2024
APRICOT, Bangkok, Thailand

A System to Detect Forge-Origin BGP Hijacks 2023
Routing Security Summit

Measuring Internet Routing from the Most Valuable Points Oct. 2022
RIPE 85, Belgrade, Serbia

An Open Platform to Teach How the Internet Practically Works Feb. 2020
NANOG 78, San Francisco, USA

Teaching and Mentoring

Lecture at Georgia Tech:

Computer Networks (CS6250) 2022

Lectures at ETH Zürich:

Communication Networks	2016 - 2022
Advanced Topics in Communication Networks	2018 - 2020
Discrete Event Systems	2019 - 2020

Students supervision:

1 Doctoral thesis supervision (Thomas Alfroy, University of Strasbourg)

3 Master thesis supervision (all at ETH Zurich)

14 Semester thesis supervision (12 at ETH Zurich, 2 at the University of Strasbourg)

Activities and Services

TPC Member for ACM IMC 2024
Reviewed five papers for ACM CoNEXT 2022
TPC Member for the poster session for ACM IMC 2022
TPC Member for the Artifact Evaluation for ACM CoNEXT 2020
Shadow TPC member for ACM IMC 2019
Occasional Reviewer for IEEE/ACM ToN and ACM CCR 2016 - 2020
CAIDA BGP Hackathon, San Diego, CA, USA (travel granted) Feb. 2016
PhD School on Software-defined Networks, Heracklion, Greece July 2015
PhD School on Traffic Monitoring and Analysis, Barcelona, Spain (travel granted) April 2015
RIPE Atlas Hackathon, Amsterdam, Netherlands, (travel granted) March 2015